Fiscal Year:	FY 2006	**Task Last Updated:**	FY 04/02/2006
PI Name:	Kaler, Eric W. Ph.D.		
Project Title:	Surfactant Structures Guide Membrane Protein Crystallization		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Other		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	kaler@che.udel.edu	**Fax:**	FY 302-831-6751
PI Organization Type:	UNIVERSITY	**Phone:**	302-831-3553
Organization Name:	University of Delaware		
PI Address 1:	Department of Chemical Engineering		
PI Address 2:	Colburn Laboratory		
PI Web Page:	http://www.kalergroup.che.udel.edu/		
City:	Newark	**State:**	DE
Zip Code:	19716	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	06/01/2001	**End Date:**	11/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Pusey, Marc	**Contact Phone:**	256 544-7823
Contact Email:	marc.pusey@msfc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lenhoff, Abraham (University of Delaware) Bahnson, Brian (University of Delaware) Naik, Ulhas (University of Delaware) Robinson, Clifford (University of Delaware)		
Grant/Contract No.:	NAG8-1830		
Performance Goal No.:			
Performance Goal Text:			
	The principal handles that we will use to influence solution behavior and crystallization are surfactants, electrolytes and hydrophilic polymers. The specific application to membrane protein crystallization will rest on characterization of protein and micellar interactions by scattering and chromatographic methods, and tuning of these interactions to guide crystallization. To reach our objective we will pursue the following specific steps: 1) Production and purification of the membrane protein of interest; 2) Measurement of interactions between protein-surfactant complexes; 3) Growth of crystals by tuning of solution conditions based on interaction measurements;		

Task Description:	4) Characterization of diffraction quality of crystals and determination of protein structure. We work with three membrane proteins, two mammalian and one bacterial. The systems are also amenable to mutation to allow investigation of the role of specific structural features in crystallization. Thus our approach is one that we expect to yield not just suitable crystals in at least a subset of the systems to be examined, but more importantly also systematic approaches for obtaining and characterizing diffraction-quality crystals of membrane proteins that can be applied to a wide variety of systems. Identification and control of the physical chemical aspects can also contribute directly to the framework needed to understand the unique features of membrane protein crystallization in microgravity. Our objective is to measure, manipulate and model the interactions among membrane proteins and their complexes with detergents, with the over-arching goal of controlling crystallization by rational manipulation of these interactions. The Protein Data Bank now contains more than 20000 structures, with more than 3000 being deposited annually, and the rate of growth is expected to expand dramatically as structural genomics efforts reach fruition. In contrast to this larger picture, only a few dozen structures of integral membrane proteins have been deposited. As a result, the structural biology of membrane proteins is much less well understood than that of soluble proteins, which is a serious shortcoming not just because of the number of membrane proteins that occur in nature, but also because of their critical role as receptors, ion channels, and membrane-bound enzymes, which makes them major targets for drug development. The availability of more structures of membrane proteins would allow rational drug design in addition to the alternative routes to discovery that must play a more important role in the absence of 3D structures. Thus there is a large potential incremental benefit of additional membrane protein structures to both the structural biology and drug design communities, and an especially large one of a fundamental understanding of membrane protein crystallization that can lead to generally applicable procedures, such as will be sought in our proposed research.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the interactions among membrane proteins and their complexes with detergents, will allow enhanced protein purification and can lead to new crystal structures of materials useful in pharaceutical applications.
Task Progress:	Previous work has shown that a correlation exists between the osmotic second virial coefficient (B22), protein solubility and crystallization for a variety of soluble proteins, and similar behavior has been observed for PDCs near a detergent phase boundary or cloud point temperature. Using self-interaction chromatography (SIC), we have measured PDC interactions for the membrane protein bacteriorhodopsin (BR) using a wide variety of precipitants and additives. A direct relationship is observed between the surfactant cloud point temperature and the corresponding PDC B22 value. This result suggests that surfactant interactions may play a constructive role in promoting PDC crystallization as well as a means by which to affect these interactions in a systematic way. Another major challenge in studying membrane proteins is identifying conditions that promote activity outside the bilayer. Results for several membrane proteins using a wide range of surfactants reveal a relationship between the surfactant hydrophile-lipophile balance (HLB) number and membrane protein activity, with a different range of optimal HLB values for each. This apparent correlation can be understood in terms of the surfactant's ability to pack efficiently around the membrane protein's transmembrane region, and is reflected in the direct relationship between surfactant HLB number and molecular packing parameter. Small-angle neutron scattering (SANS) experiments were carried out on protein-surfactant complexes of bacteriorhodopsin (a seven-helical integral membrane protein from Halobacterium salinarium which acts as a light driven proton pump) in mixtures of n-octyl-ß-D-glucoside (C8ßG1) and triethylene glycol mono-n-octyl ether (C8E3). A model was developed to decouple particle form factor and structure factor of the protein-surfactant complexes, once the microstructure of the empty micelles was characterized. Results show that the interactions between protein-surfactant complexes become progressively more attractive as the surfactant mixture becomes richer in C8E3, thus following the same pattern as the protein-free surfactant solutions, which were also measured. These results prove that for bacteriorhodopsin the major component to the interactions among complexes is due to the surfactant bound to the hydrophobic part of the protein. We also extended the characterization of surfactant solutions for potential use in membrane protein crystallization. In particular, we found interesting patterns of phase behavior for short chain alkyl monoglucoside surfactants (C8ßG1 and C9ßG1) in solutions with polyethylene glycol (PEG) and ammonium sulfate, which are commonly used as precipitants in crystallization screens. These surfactants are soluble in aqueous solutions of ammonium sulfate up to 4 M ionic strength. In combination with PEG of molecular weight from 3350 to 10000, two phase transitions can be observed, with lower and upper two-phase regions separated by a single isotropic solution. The lower miscibility region becomes larger with increasing the PEG molecular weight or the surfactant hydrophobic tail. SANS experiments were used to characterize the microstructure of these solutions in the one-phase regions near the lower miscibility gap. The scattering data were evaluated with the Generalized Indirect Fourier Transformation (GIFT) technique, which allowed separation of information about the size and shape of the glucoside micelles and their interactions in solutions with different ratios of PEG 3350. We observed that addition of PEG significantly affects both these features, and for solutions of octyl glucoside it leads to weakly attractive interactions in proximity of the phase boundary. This explains the increasing success of this specific combination of additive and surfactant in crystallization screens. Alkyl polyglucosides are the most used surfactants in extraction and stabilization of membrane proteins. Self-assembly of the integral membrane protein bacteriorhodopsin and various alkyl polyglucosides (monoglucosides and maltosides) was studied by sedimentation equilibrium and small-angle neutron scattering experiments. By using the absorption characteristic of the retinal chromophore bound to the active form of bacteriorhodopsin, the features of the surfactant-bound protein monomers could be successfully characterized. In the case of the alkyl monoglucoside C8ßG1 and C9ßG1 the amount of surfactant bound to bacteriorhodopsin was confirmed in parallel small-angle neutron scattering experiments. Contrast variation studies were performed with solutions at different levels of deuteration to highlight the properties of the bound surfactant. Results from both experiments were analyzed in view of the stability of bacteriorhodopsin in these surfactants measured over the course of several days. Results show that optimal stability occurs when the number of surfactant molecules bound is approximately equal to or slightly lower than the micellar aggregation number. Situations where the bound surfactant exceeds the micellar number (C8ßG2 and C9ßG2) correlate with limited stability of bacteriorhodopsin. The effect of additives commonly used to control interaction between protein-surfactant complexes, such as salt and the water soluble polyethylene glycol (PEG) on the binding properties of alkyl polyglucosides was also investigated. Results show that PEG does not bind to the complexes, whereas addition of 0.5 M ammonium sulfate does increase the amount of bound surfactant as well as the micellar aggregation number. These observations are especially relevant for understanding the mechanism of membrane protein crystallization in presence of these additives.

Bibliography Type:	Description: (Last Updated: 03/17/2017)
Articles in Peer-reviewed Journals	Berger, B.W.; Blamey, C.; Naik U.; Bahnson B.J.; Lenhoff A.M. "Roles of Additives and Precipitants in Crystallization of Calcium- and Integrin-Binding Protein." Crystal Growth and Design 2005 Jul;5(4): 1499-507. , Jul-2005
Articles in Peer-reviewed Journals	Berger BW, Gendron CM, Robinson CR, Kaler EW, Lenhoff AM. "The Role of Protein and Surfactant Interactions in Membrane Protein Crystallization." Acta Crystallogr D Biol Crystallogr. 2005 Jun;61(Pt 6):724-30. PMID: 15930629 , Jun-2005
Articles in Peer-reviewed Journals	Berger BW, Garcia RY, Lenhoff AM, Kaler EW, Robinson CR. "Relating Surfactant Properties to Activity and Stability of Human Adenosine A3 Receptor." Biophys J. 2005 Jul;89(1):452-64. PMID: 15849244 , Jul-2005
Articles in Peer-reviewed Journals	Santonicola GM, Kaler EW. "Mixtures of n-Octyl-ß-D-glucoside and Triethylene Glycol Mono-n-octyl Ether: Phase Behavior and Micellar Structure near the Liquid-Liquid Phase Boundary. " Langmuir. 2005 Oct 25;21(22):9955-63. PMID: 16229514 , Oct-2005